CSE221: Algorithms
Lab-01
Fall 2026

Action required for Students

[Google Form] CSE221 Lab Policy:
- The lab policy file can be found here - Link
- Please complete the lab policy acknowledgement form - Link

[Google Form] Codeforces Account Information: (Action required)
- Open an account at https://codeforces.com/
- Please submit your codeforces details here - Link

Students who do not complete this will not be allowed to participate in lab quizzes, and their assignment submissions will not be graded.

Announcements for Students

[Bookmark the Link] Student Drive Folder: Link
The folder contains:
- Lab Files [Fall 2026]
- Lab Policy
- Sample Quiz Checkers
- Environment Setup for Quiz Checker
- Previous Semesters' Lab Quiz Questions

Graded Quiz 01
Date: October 17-18, 2026 (as per your lab class schedule)
Syllabus: Lab 01 content
Take Preparation for both assignment problems and outside assignment problems *[Student Drive Folder Link]*

Assignment-01 Links:

Codeforces Invitation Link:
https://codeforces.com/contestInvitation/8bf630eb2ce286ebc86d1dfbf93a8d8fbec39e83

Alternative Link: https://codeforces.com/contests/722945
[The alternative link will **only work** after you have successfully registered using the first link.]

Self-study: Explore Fast I/O techniques: We must use FAST I/O in our code.

- **Python**: Using `sys.stdin` for faster input handling.
- **C++**: Using `ios_base::sync_with_stdio(false);` and `cin.tie(nullptr)`
- **Java**: Using `BufferedReader` for efficient input reading.
- The fast i/o can reduce program execution time on Online Judge platforms.
- **Fast I/O Example: Python, C++, Java [click Expand to view the code snippet]**

Deadline: `Oct 18, 2026 2:00 PM`
After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading. Also, your assignment submissions will not be evaluated if they are flagged as AI-generated.**

Learning Outcome of this Lab:

- Understand the basics of Online Judge platforms.
- Become familiar with standard Input/Output (I/O) formats used in Online judge
- Learn about common judge verdicts, including Accepted (AC), Wrong Answer (WA), Time Limit Exceeded (TLE), and Runtime Error (RE).
    - For a **1-second** time limit, most OJs allow approximately **10^8 operations**.
    - For python select PyPy over Python if available, as PyPy is faster. However, PyPy consumes more memory than Python.

- Review $O(n^2)$ sorting algorithms which have already been covered in the previous courses such as Bubble Sort, Selection Sort, and Insertion Sort.
    - Learn to sort with multiple parameter
    - Understand the use case of stable sort

CSE221 Assignment 01 Fall 2026

A. Odd or Even?

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Do you know how to tell if a number is **Odd** or **Even**? You are given T numbers, and for each of those numbers, you have to tell whether the number is odd or even.

Input
The first line will contain a single integer T ($1 \leq T \leq 100$). Each of the next T lines will contain a number N ($-10^5 \leq N \leq 10^5$).

Output
For each N, you have to print whether the number is odd or even. See the sample input-output format to know what exactly you have to print.

input

```
5
10
19
7
3
100
```

output

```
10 is an Even number.
19 is an Odd number.
7 is an Odd number.
3 is an Odd number.
100 is an Even number.
```

B. Calculator

1 second, 256 megabytes

Forbidden Keywords for the Quiz: eval, sort, open, file, creat(, fstream, thread, process, system(, exec(

In this problem, you have to implement a mini-calculator.

Input
The first line will contain an integer T ($1 \leq T \leq 1000$) representing the number of test cases.

For each test case, there will be a line containing the world `calculate` followed by a simple arithmetic operation of the form a `OP` b, where a and b are non-negative integers $0 \leq a, b \leq 1000$ and `OP` is one of `+`, `-`, `*`, `/`, `%`, `&`, `|`, `^`, $\ll$, or $\gg$, where these operators mean addition, subtraction, multiplication, division, modulo, bitwise AND, bitwise OR, bitwise XOR, left shift, and right shift respectively.

Output
For each test case, you have to print the result. It is guaranteed that the result will be between -2^{63} and $2^{63} - 1$. Your answer will be considered correct if its absolute or relative error is within 10^{-6} of the correct result. More formally, if your answer is a and the correct answer is b, it will be considered correct if and only if $|a - b| \leq 10^{-6}$ or $\frac{|a-b|}{|b|} \leq 10^{-6}$.

Note on Floating Point Output: Your answer may contain floating point numbers. In that case, it does not have to be exactly equal to the correct answer. As long as the absolute difference or relative error between your answer and the correct answer is at most 10^{-6}, it will be accepted. For example, if your answer is 20.250000001 and the correct answer is 20.25, it will be accepted. Similarly, if your answer is 19 and the correct answer is 19.0000000000, it will also be accepted.

input

```
11
calculate 1 + 3
calculate 5 - 2
calculate 7 * 5
calculate 8 / 4
calculate 9 / 4
calculate 9 % 4
calculate 5 & 6
calculate 5 | 6
calculate 5 ^ 6
calculate 5 << 1
calculate 5 >> 1
```

output

```
4
3
35
2.0000000000
2.2500000000
1
4
7
3
10
2
```

input

```
4
calculate 749 / 291
calculate 419 / 226
calculate 1000 << 53
calculate 768 << 49
```

output

```
2.5738831615
1.8539823009
9007199254740992000
432345564227567616
```

Please note that except for the division, all other operations (especially the bitwise operations `&`, `|`, and `^`) are integer/long operations.

C. Fast Sum

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Your friend is trying to solve the following problem. You are given $\mathbf{T}$ test cases. For each test case, you are given an integer $\mathbf{N}$. You have to find out the summation of 1 to N. More formally, your friend has to calcuate

$$\sum_{x=1}^{x=N} x$$

Your friend wrote the following code in Python to solve it:

```python
T = int(input())

for _ in range(T):
    N = int(input())
    sum = 0
    for i in range(1, N + 1):
        sum += i
    print(sum)
```

Same code in Java:

```java
import java.util.Scanner;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int T = sc.nextInt();
        for (int t = 0; t < T; t++) {
            int N = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= N; i++)
                sum += i;
            System.out.println(sum);
        }
    }
}
```

However, the code is not passing the online judge due to some unknown errors for large values of N.

Since you are currently studying CSE221 and have learned about time complexity, help your friend come up with a more efficient solution.

Input
The first line contains a single integer T $(1 \leq T \leq 10^4)$ — the number of test cases.

The next T lines each contain a single integer N $(1 \leq N \leq 10^6)$.

Output
For each test case, print a single integer — the summation from 1 to N.

input
5 2 5 10 12 100
output
3 15 55 78 5050

D. Is Sorted?

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

You are given an array of N integers. Determine whether the given array is in non-decreasing order.

An array is said to be in non-decreasing order if, for every valid index i such that $1 \leq i < N$ (1-based indexing), the condition $A[i] \leq A[i+1]$ holds true.

Example: $[1, 2, 4, 5]$, $[1, 2, 2, 4, 4, 5]$ are in the non-decreasing order because every element is less than or equal to the one after it.

Input
The first line contains a single integer T $(1 \leq T \leq 100)$ — the number of test cases. Each testcase contains two lines.

In each test case, the first line contains a single integer N $(1 \leq N \leq 10^4)$ — the number of elements in the array. The second line contains N integers separated by spaces $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq 10^6)$ — the elements of the array.

Output
For each test case, print YES if the array is in non-decreasing order. Otherwise, print NO.

input
3 4 1 2 3 3 4 1 5 2 6 1 5
output
YES NO YES

E. Reverse Sorting

2 seconds, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

You are given an array A of $\mathbf{N}$ integers. Your task is to sort the array in non-decreasing order using only a specific type of operation:

- In one operation, you may select any subarray of length exactly 3 and reverse it. In other words, you choose some $1 \leq i, j \leq N$ such that $j - i + 1 = 3$ and reverse `A[i...j]`.

You can apply this operation as many times as you like (or not at all). Your goal is to determine whether it is possible to sort the array using only this operation. If it is possible, you have to print the operations too. Look at the output format for a better understanding.

Input
The first line contains an integer N $(1 \leq N \leq 1000)$. The second line contains N integers A_i $(1 \leq S_i \leq 10^5)$.

Output
If it is not possible to make the array non-decreasing, print **NO**. Otherwise print **YES**. If your answer is **YES**, you have to output the number of moves you needed, let it be M. Then the next M lines will contain pairs of (i, j) representing a valid operation. Make sure your moves are valid, and those moves make the array A non-decreasing.

NOTE: If you don't understand the output format properly, look at the sample input-output and explanation.

input
4 2 3 1 1
output
YES 2 1 3 2 4

input
6 2 5 5 1 5 5
output
NO

input
1 2
output
YES 0

input
2 6 6
output
YES 0

For the Sample Input, one way to sort the array $[2, 3, 1, 1]$ using only the allowed operation:

1. Choose indices $(1, 3)$ and reverse the subarray $[2, 3, 1]$ which becomes $[1, 3, 2]$. So the resulting array: $[1, 3, 2, 1]$
2. Choose indices $(2, 4)$ and reverse subarray $[3, 2, 1]$ which becomes $[1, 2, 3]$. So the resulting array: $[1, 1, 2, 3]$

F. An Ancient Sorting Algorithm

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

You are given an array of N integers. You have to sort the array in non-decreasing order using a custom sorting algorithm with the following constraint:

- You may only swap adjacent elements with the same parity (i.e., both even or both odd)

Sort the array in non-decreasing order until no more such swaps are possible and print the final array.

Input
The first line contains a single integer N $(1 \leq N \leq 1000)$ — the number of elements in the array.

The second line contains N integers separated by spaces $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq 10^6)$ — the elements of the array.

Output
Print the final array after sorting the array in non-decreasing order until no more such swaps are possible.

input
7 4 2 4 7 1 6 1
output
2 4 4 1 7 6 1

input
5 3 5 9 7 1
output
1 3 5 7 9

input
14 4 8 2 9 1 5 4 6 8 1 7 13 11 8
output
2 4 8 1 5 9 4 6 8 1 7 11 13 8

input
1 221
output
221

G. Sorting Again??

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Suppose you are given a task to rank the students. You have gotten the marks and ID of the students. Now your task is to rank the students based on their marks using a sorting algorithm. If two or more students get the same mark, then students with the lower ID will get prioritized. See the input and output for a better understanding.

However, **you have to keep in mind that your sorting algorithms perform the minimum number of swapping operations.**

Input
The first line contains an integer T $(1 \leq T \leq 100)$ denoting the number of test cases.

For each test case:

The first line contains an integer N $(1 \leq N \leq 1000)$.

The second line contains N distinct integers S_i $(1 \leq S_i \leq 1000)$, where S_i denotes the student ID.

The third line contains N integers S_m $(1 \leq S_m \leq 1000)$, where S_m denotes the obtained mark of the corresponding student.

Note: It is guaranteed that all student IDs are unique.

Output
For each test case, the output must contain a number X which denotes the number of minimum swaps. The rest of the N lines will contain the Student ID and obtained marks sorted based on the instruction above. See the sample output for a better understanding.

Important Note: Since you are asked to minimize the number of swaps, if your number of swaps doesn't match with the judge's answer, your solution will be considered incorrect.

TO CHANGE: Look at the first testcase of the sample input. It can be shown that this can be sorted with only 4 swaps. It can also be shown that it is not possible to sort this in less than 4 swaps.

input
2 7 7 4 9 3 2 5 1 40 50 50 20 10 10 10 3 1 2 3 40 59 10
output
Minimum swaps: 4 ID: 4 Mark: 50 ID: 9 Mark: 50 ID: 7 Mark: 40 ID: 3 Mark: 20 ID: 1 Mark: 10 ID: 2 Mark: 10 ID: 5 Mark: 10 Minimum swaps: 1 ID: 2 Mark: 59 ID: 1 Mark: 40 ID: 3 Mark: 10

H. Trains?

3 seconds, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You have been recently recruited as the Software Engineer at Jumanji Railway Software System. You have a big task at hand. You will be given $N(1 \leq N \leq 100)$ schedule of the train. The next N line will contain the name of the train and the departure time. See the input format for better understanding.

Your task is to write a sorting algorithm that will group the trains in the lexicographical order based on the name of the trains. If two or more trains have the same name, then the train with the latest departure time will get prioritized. If there is still a tie, then the train which comes first in the input will come first.

Note: You may use any built-in sorting function for this problem.

Input

The first line will contain an integer $N(1 \leq N \leq 100)$. For the next N lines, i_{th} line will describe i_{th} train. Please see the sample input for better understanding.

Please note that the names of the trains and destinations don't contain any white spaces, and the length of the names and destinations will be at most 100. For example, look at the following description:

```
DhumketuExpress will departure for Chittagong at 02:30
```

Here, **DhumketiExpress** is the name of the train **Chittagong** is the destination, and they don't contain any whitespaces, and their length is less than 100.

Output

Print the train description in the sorted order (specified above). Please see the output format for better understanding.

input
8 ABCD will departure for Mymensingh at 00:30 DhumketuExpress will departure for Chittagong at 02:30 ABC will departure for Dhaka at 17:30 ABCD will departure for Chittagong at 01:00 ABC will departure for Khulna at 03:00 ABC will departure for Barisal at 03:00 ABCE will departure for Sylhet at 23:05 PadmaExpress will departure for Dhaka at 19:30
output
ABC will departure for Dhaka at 17:30 ABC will departure for Khulna at 03:00 ABC will departure for Barisal at 03:00 ABCD will departure for Chittagong at 01:00 ABCD will departure for Mymensingh at 00:30 ABCE will departure for Sylhet at 23:05 DhumketuExpress will departure for Chittagong at 02:30 PadmaExpress will departure for Dhaka at 19:30

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